

5 Quick sort

08 March 2025 19:31

→ smallest person comes automatically to first and biggest to last and rest automatically take place

→ element is in sorted position when left side of elements are smaller than element and right side of elements are bigger than element

ex (10), 80, 90, 60, 20

5, 3, 2, 1 (1)

46, 7, (10) 16, 12, 13, 14

Circle elements are at sorted position

→ its Divide and Conquer Sort

→ A

10	16	8	12	15	6	3	9	5	∞
----	----	---	----	----	---	---	---	---	---

 $\xrightarrow{i}$ $\xleftarrow{j}$ $\xrightarrow{h}$ infiny (highest number)

pivot = 10

i find number greater than pivot
j find number smaller than pivot

10	16	8	12	15	6	3	9	5	∞
----	----	---	----	----	---	---	---	---	---

↑ i ← j

10	5	8	12	15	6	3	9	16	∞
----	---	---	----	----	---	---	---	----	---

i ← j

10	5	8	9	15	6	3	12	16	∞
----	---	---	---	----	---	---	----	----	---

i ← j

10	5	8	9	3	6	15	12	16	∞
----	---	---	---	---	---	----	----	----	---

i ← j

6	5	8	9	3	10	15	12	16	∞
---	---	---	---	---	----	----	----	----	---

at its sorted position

partition(l, h)

{ pivot = A[l];
i = l; j = h;

```

while (i < j)
{
    do {
        i++;
    } while (A[i] ≤ pivot);

    do {
        j--;
    } while (A[j] > pivot);

    if (i < j)
    {
        swap (A[i], A[j]);
    }

    swap (A[l], A[j]);

    return j;
}

```

```

Quick Sort (l, h)
{
    if (l < h)
    {
        j = partition(l, h);
        Quick Sort (l, j);
        Quick Sort (j+1, h);
    }
}

```